

Hope Artificial Intelligence

Scenario Based Learning

A company works with number of employees; all the works are dependents on the employees. Even if one of the employees resign the job immediately then assigned work will be not finished at the time, so delivery of the project to the clients will be delayed. Company planned to make solution for this, they want to know which employee may resign next. If they know previously, they can arrange alternative to avoid such problem. As an AI Engineer you must give Solution to this.

- A) How will you achieve this in AI?
- B) Find out the 3 -Stage of Problem Identification
- C) Name the project
- D) Create the dummy Dataset.

Dummy dataset:

S.No.	Emp. Name	Emp. Code	DOJ	Designation	Grade	Role	Project 1	Performance on project 1 - 5	Project 2	Performance on project 1 - 5	Annual Leave 24days	Behaviourial Performance 1 - 5	Overall appraisal rating 1 - 5	Exp.in years	Resigned Yes/No
1	Sundar M	100451	21-Aug-20	VP	7	Tower Head	MainStream	5	FinLea	5	20	5	5.0	5.7	No
2	Ramesh M	102519	17-Apr-21	AVP	6	Project Head	AvivaStocks	5	SwisTek	4	22	4	4.3	5.0	No
3	Vimalkumar S	104348	5-Jun-21	Project Manager	4	FullStack	FinLea	5	AvivaStocks	5	26	4	4.7	4.9	No
4	Pragadheesh S	106128	11-Mar-22	Project Manager	4	Architect	SwisTek	3	FinLea	3	25	3	3.0	4.1	No
5	Magesh T	107528	10-Sep-22	Project Manager	4	SWE	AvivaStocks	4	SwisTek	4	20	4	4.0	3.6	No
6	Deepak V	107629	20-Apr-22	Jr.Consultant	1	SWE	AvivaStocks	3	FinLea	3	30	2	2.7	4.0	Yes
7	Kamesh V	109090	5-May-22	Sr.Consultant	3	SWE	AvivaStocks	5	SwisTek	5	21	4	4.7	4.0	No
8	Vijayakumar A	110536	2-Nov-22	Jr.Consultant	1	FullStack	FinLea	4	AvivaStocks	4	20	4	4.0	3.5	No
9	Chandrika K	111801	13-Feb-25	Sr.Consultant	2	FullStack	FinLea	2.5	AvivaStocks	2.5	29	2.5	2.5	1.2	Yes
10	Aadhikesavan H	112963	1-Jan-24	Jr.Consultant	1	FullStack	FinLea	2	AvivaStocks	3	29	2	2.3	2.3	Yes
11	Dhanasekar B	113803	15-Jan-25	Sr.Consultant	2	Architect	SwisTek	3	FinLea	4	22	3.5	3.5	1.3	Yes
12	Bharathkumar J	114629	19-Feb-25	Jr.Consultant	1	Architect	SwisTek	3.5	FinLea	3.5	24	3.5	3.5	1.2	Yes
13	Priyanka V	115054	24-Jan-25	Sr.Consultant	2	SWE	MainStream	3.5	SwisTek	2.5	23	3.5	3.2	1.2	Yes

Stage 1: Identify Domain:

Here we have more numerical values are defines the performance. Hence the Machine Learning will be suitable.

Stage 2: Check input & output:

Secondly both input and output fulfils; Hence the Supervised Learning will be suitable.

Stage 3: Choose suitable model:

The output data is in categorical values hence the classification will be suitable.

Logic:

Problem statement here we need to predict the upcoming resigning employees.

Check the last appraisal rating of employees and tenure they are served in the company.

Upon checked the employees were rated less than 3 in last appraisal AND as they served less than 3 years in the company.

Those who were resigned in previously and the above two conditions categorised as Yes/No.

In addition, we also have another parameter on “Behaviour Performance Rating” for weightage to predict those who were resigned in previously.

Solution:

Categorical data provided the clear data about the resigned employees.

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AI Predication:

If the current employee would be rated in less than 3.

And the employee serves less than 3 years.

These conditions for the current employees lead to attrition.

AI prediction tool will highlight in red category employee with help of supervised learning.